AIRWATCH INDIA: AQI INSPECTION AND PREDICTION

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Term Project Report
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ABSTRACT

Air pollution is a persistent public-health concern across Indian cities, yet accessible tools that combine historical inspection with short-term forecasting remain limited. This report presents an end-to-end platform for Air Quality Index (AQI) inspection and prediction covering 26 Indian cities from 2015–2020. Daily pollutant concentrations sourced from the Central Pollution Control Board [1] were cleaned, and 4,680 missing AQI values were recovered by recomputing the official CPCB sub-index formula [2], yielding a final dataset of 29,530 records with zero missing values. Exploratory analysis revealed a strong winter pollution peak (November average AQI 234.9 versus August's 110.0) and identified carbon monoxide and PM2.5 as the pollutants most correlated with AQI ($r = 0.67$ and $r = 0.66$). A Random Forest regressor predicted daily AQI with $R^2 = 0.925$ and a Random Forest classifier predicted the six-tier AQI category with 83.8% accuracy. Holt-Winters exponential smoothing forecast six-month-ahead AQI for the ten most polluted cities with sufficient historical depth. All findings are exposed through a four-tab interactive Streamlit dashboard supporting live filtering, city inspection, on-demand prediction, and geographic visualization.

Index Terms— Air Quality Index, CPCB, Machine Learning, Random Forest, Time-Series Forecasting, Streamlit Dashboard, India

1. INTRODUCTION

Air pollution is among the leading environmental health risks in India, with several cities regularly exceeding World Health Organization guideline limits for particulate matter [3]. Public authorities publish raw pollutant readings, but tools that integrate historical inspection, statistical analysis, and short-term prediction into a single accessible interface remain scarce for the general public and student researchers alike.

This project addresses that gap by building a complete data science pipeline applied to daily air quality records for 26 geographically diverse Indian cities. The pipeline spans data cleaning and AQI recomputation, exploratory and correlation analysis, feature engineering, classical time-series forecasting, supervised machine learning for both regression and classification, and an interactive web dashboard, implemented entirely in Python using `pandas` [4], `scikit-learn` [5], and `statsmodels` [6].

Section 2 defines the problem and objectives. Section 3 details the methodology. Section 4 presents experimental results. Section 5 discusses findings and limitations. Section 6 concludes with future directions, and Section 7 lists artifacts and demonstration links.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

The problem addressed is the absence of an integrated, reproducible platform for air quality inspection and short-term prediction specific to Indian cities. Publicly available datasets contain substantial missing values, including in the AQI field itself, and existing dashboards rarely combine historical inspection, machine-learning-based prediction, and geographic visualization in one open-source tool. The scope covers eleven pollutant variables (PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene) for 26 cities across 21 states over 5.5 years (January 2015 – July 2020).

2.2. Objectives

1. **Collect** daily city-level pollutant and AQI records from the Central Pollution Control Board via `data.gov.in` [1].
2. **Clean and recover** missing values, including recomputing missing AQI entries using the official CPCB sub-index formula [2].
3. **Analyse** seasonal patterns, city/state-wise pollution levels, and pollutant–AQI correlations.
4. **Engineer** temporal, seasonal, and lag/rolling features to support forecasting and prediction.
5. **Forecast** six-month-ahead AQI for the most polluted cities using Holt-Winters exponential smoothing [7].
6. **Train and evaluate** regression and classification models to predict AQI value and category from pollutant readings.
7. **Deploy** an interactive Streamlit dashboard for live inspection, prediction, and map-based exploration.

3. METHODOLOGY / APPROACH

3.1. Overall Workflow

The system was structured as a nine-phase sequential pipeline, each producing verifiable artefacts reviewed before proceeding:

1. **Data cleaning:** load raw CPCB records, merge station metadata, impute pollutants, recompute missing AQI.
2. **Exploratory analysis:** trend, seasonal, city/state, and correlation analysis with statistical summaries.
3. **Feature engineering:** temporal, seasonal, and lag/rolling AQI features.
4. **Geographic visualization:** interactive Folium maps of city-wise average AQI.

5. **Time-series forecasting:** seasonal decomposition and Holt-Winters forecasting for high-pollution cities.
6. **Machine learning:** regression and classification model training and comparison.
7. **Model evaluation:** metrics tables, confusion matrix, and feature-importance analysis.
8. **Dashboard development:** a four-tab Streamlit application unifying all prior outputs.
9. **Reporting:** consolidation of findings into this report and accompanying artefacts.

This staged design allowed each phase to be independently verified against expected outputs before downstream phases consumed its results, reducing the risk of silently propagating data errors.

3.2. Data Cleaning and AQI Recovery

The raw dataset comprised 29,531 daily records across 26 cities. The Benzene-family pollutant Xylene was dropped (61% missing). Remaining pollutant gaps were imputed using city-and-month-wise medians, with city-level and global medians as fallback. Critically, 15.9% of AQI values (4,681 rows) were missing. Rather than discard these rows, each was recomputed using the official CPCB sub-index formula [2]: for a pollutant concentration C falling in breakpoint range $[C_{\mathrm{lo}}, C_{\mathrm{hi}}]$ mapping to sub-index range $[I_{\mathrm{lo}}, I_{\mathrm{hi}}]$,

$$I = \frac{I_{\mathrm{hi}} - I_{\mathrm{lo}}}{C_{\mathrm{hi}} - C_{\mathrm{lo}}} (C - C_{\mathrm{lo}}) + I_{\mathrm{lo}}. \tag{1}$$

The overall AQI is the maximum sub-index across available pollutants, computed only when at least three pollutants including PM2.5 or PM10 were present, per CPCB convention. This recovered 4,680 of 4,681 missing rows (99.98%), leaving a final dataset of 29,530 records with zero missing values across all fields. Table 1 lists the resulting six-tier CPCB category scheme used as the classification target throughout this project.

Table 1. CPCB AQI category breakpoints.

Category	AQI Range
Good	0–50
Satisfactory	51–100
Moderate	101–200
Poor	201–300
Very Poor	301–400
Severe	401+

3.3. Feature Engineering

Date-derived features (year, month, day, day-of-week) and an Indian seasonal mapping (Winter, Summer, Monsoon, Post-Monsoon) were added. Per-city lag features (previous day and previous week AQI) and rolling averages (7-day and 30-day, computed on lagged values to avoid leakage) were engineered to support both forecasting and prediction tasks. City and state identifiers were label-encoded.

3.4. Time-Series Forecasting

For the ten most polluted cities with at least 24 months of history (required for valid seasonal decomposition), monthly-resampled AQI series were decomposed into trend, seasonal, and residual components, then forecast six months ahead using Holt-Winters exponential smoothing [7]:

$$\ell_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}) \tag{2}$$
$$b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1} \tag{3}$$
$$s_t = \gamma(y_t - \ell_t) + (1-\gamma)s_{t-m} \tag{4}$$

where ℓ_t, b_t, and s_t are the level, trend, and seasonal components at time t, $m = 12$ is the seasonal period, and $\alpha, \beta, \gamma \in [0, 1]$ are smoothing parameters fitted by maximum likelihood.

3.5. Machine Learning Models

Two supervised tasks were trained on an 80/20 train-test split (random state fixed at 42) using pollutant concentrations, temporal features, and lag/rolling AQI features: (i) *regression* to predict continuous AQI using Linear Regression, Decision Tree, and Random Forest [8] regressors; and (ii) *classification* to predict the six-tier AQI category (Good, Satisfactory, Moderate, Poor, Very Poor, Severe) using Logistic Regression, Decision Tree, and Random Forest classifiers, implemented with `scikit-learn` [5].

3.6. Dashboard Architecture

Findings were exposed through a four-tab Streamlit dashboard: an *Overview* tab with national KPIs and trend charts; a *City Inspection* tab with per-city AQI trends, seasonal patterns, and pollutant breakdowns; a *Predict AQI* tab that trains the Random Forest models once (cached at startup) and accepts live pollutant inputs to return a predicted AQI value, category, and gauge visualization; and a *Map View* tab embedding the pre-generated Folium map. A sidebar exposes live filters (state, city, category, year range, minimum AQI) that propagate to the Overview and City Inspection tabs without recomputation of the underlying models.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

All experiments were conducted on Ubuntu 24 with Python 3.12. Key libraries: `pandas`, `scikit-learn`, `statsmodels`, `matplotlib`, `seaborn`, `plotly`, and `folium`. Models were trained on 23,624 records and evaluated on 5,906 held-out records.

4.2. Exploratory Findings

The dataset spans 2015-01-01 to 2020-07-01, with an overall average AQI of 162.2. Ahmedabad recorded the highest average AQI (392.6, “Very Poor”), while Aizawl recorded the lowest (34.8, “Good”). A pronounced seasonal cycle emerged: November averaged 234.9 versus August’s 110.0, consistent with winter temperature inversion trapping pollutants near ground level. Figure 1 shows the national monthly AQI trend, with a clear annual winter peak repeating across all five years.

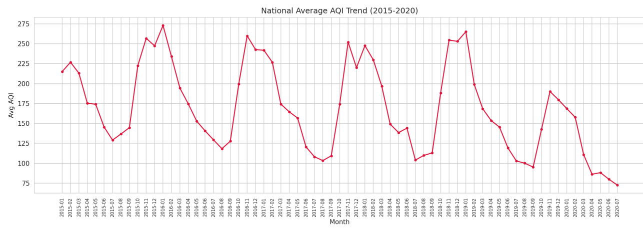

Fig. 1. National average monthly AQI (2015–2020), showing a recurring winter peak and monsoon-season minimum each year.

Figure 2 shows the pollutant correlation heatmap. Carbon monoxide ($r = 0.67$) and PM2.5 ($r = 0.66$) are most strongly correlated with AQI, followed by NO2 ($r = 0.53$), consistent with their role as primary combustion tracers.

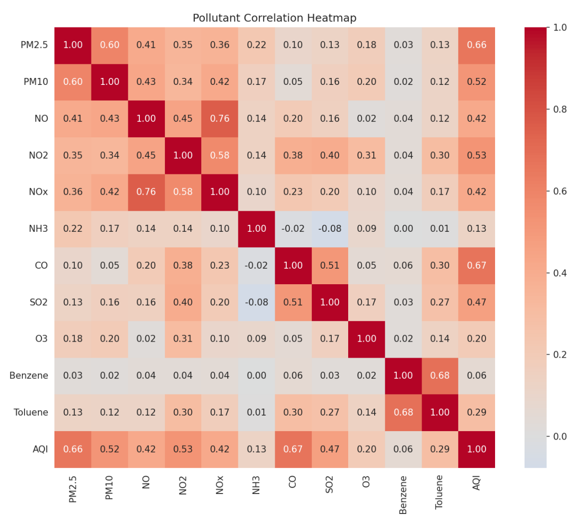

Fig. 2. Pollutant correlation heatmap. CO and PM2.5 show the strongest linear association with AQI.

Table 2 summarizes the five most and least polluted cities by average AQI.

Table 2. Top and bottom five cities by average AQI (2015–2020).

Most Polluted	**Avg AQI**	**Least Polluted**	**Avg AQI**
Ahmedabad	392.6	Aizawl	34.8
Delhi	258.7	Shillong	54.6
Gurugram	226.6	Coimbatore	73.4
Lucknow	219.7	Thiruvananthapuram	73.9
Patna	216.6	Ernakulam	92.2

4.3. Forecasting Results

Holt-Winters models converged for all ten eligible cities. For Ahmedabad, the forecast correctly projected the seasonal winter peak, estimating November AQI at 366.3, consistent with the city's historical November average. Cities with under 24 months of history (e.g., Guwahati) were excluded from decomposition and replaced by the next most-polluted eligible city (Jaipur) to preserve statistical validity.

4.4. Machine Learning Results

Table 3 summarizes model performance. Random Forest achieved the best results on both tasks: $R^2 = 0.925$ (MAE 16.33, RMSE 35.37) for AQI regression, and 83.8% accuracy (F1 0.837) for AQI category classification.

Table 3. Model performance on held-out test data (20% split).

Regression Model	R^2	MAE	RMSE
Linear Regression	0.897	21.84	41.37
Decision Tree	0.865	20.75	47.33
Random Forest	**0.925**	**16.33**	**35.37**
Classification Model	Accuracy	F1	–
Logistic Regression	0.797	0.794	–
Decision Tree	0.823	0.822	–
Random Forest	**0.838**	**0.837**	–

Figure 3 shows actual versus predicted AQI for the Random Forest regressor, with predictions closely tracking the identity line across the bulk of the AQI range, with modest divergence at extreme values (AQI > 1000).

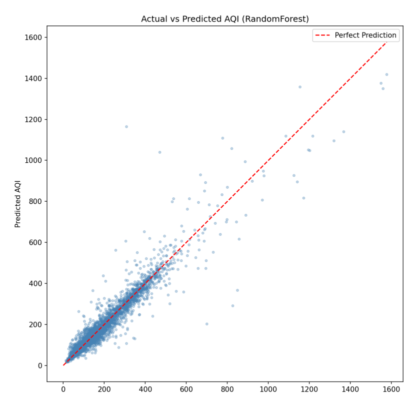

Fig. 3. Actual vs. predicted AQI for the Random Forest regressor on held-out test data.

Feature-importance analysis of the Random Forest regressor identified the previous-day AQI lag feature as by far the most influential predictor, followed distantly by CO and PM2.5 concentrations; date parts and one-hot season indicators contributed negligibly once lag and pollutant features were available, confirming that short-term persistence combined with combustion tracers explains most day-to-day AQI variation. For the classification task, the confusion matrix showed that the majority of errors occurred between adjacent categories (Satisfactory/Moderate and Moderate/Poor), while the Good and Severe extremes were classified with high precision.

5. DISCUSSION

Combustion tracers dominate. CO and PM2.5's correlation with AQI confirms combustion-related pollution as the primary driver of poor air quality, rather than secondary pollutants such as ozone.

Winter inversion and regional disparity. The consistent November–December peak reflects atmospheric temperature inversion during Indian winters. Gujarat, Delhi, Haryana, Uttar Pradesh, and Bihar recorded the five highest average AQI values, while northeastern and southern coastal states recorded the lowest.

AQI recomputation validity. Recovering 99.98% of missing AQI values via the CPCB formula improved completeness while remaining faithful to the official standard.

Limitations. Random Forest's accuracy partly reflects the near-deterministic relationship between same-day pollutant readings and AQI; misclassifications concentrated between adjacent categories; and cities with short monitoring histories were excluded from decomposition-based forecasting.

6. CONCLUSION AND FUTURE WORK

This project demonstrates a complete, reproducible pipeline for air quality inspection and prediction across 26 Indian cities. Starting from a raw CPCB dataset with substantial missing data, the platform delivers a cleaned 29,530-record dataset with zero missing values, 13 analysis figures, three interactive geographic maps, a Random Forest regressor ($R^2 = 0.925$) and classifier (83.8% accuracy), six-month Holt-Winters forecasts for ten cities, and a live four-tab Streamlit dashboard. Three conclusions emerge: (i) combustion-related pollutants (CO, PM2.5) are the dominant drivers of AQI variation, rather than secondary photochemical pollutants such as ozone; (ii) a strong, recurring winter pollution peak driven by atmospheric temperature inversion is detectable nationwide; and (iii) ensemble tree-based models substantially outperform linear baselines for both AQI regression and category classification.

Future work should address four directions:

1. Extend coverage to real-time CPCB station APIs for live inspection beyond the 2015–2020 historical window.
2. Incorporate meteorological covariates (temperature, wind speed, humidity) as exogenous regressors to improve forecast skill.
3. Explore SARIMA or hybrid statistical-ML models to quantify the accuracy trade-off against Holt-Winters on this dataset.
4. Deploy the Streamlit dashboard to a public cloud host for access without local installation.

7. ARTIFACTS AND DEMONSTRATIONS

- **GitHub Repo:** `https://github.com/vjnitin312/da379` — scripts, cleaned dataset, 13 figures, 3 Folium maps, and README with setup instructions.
- **Dashboard:** `streamlit run dashboard/app.py` after cloning — four-tab interactive dashboard (Overview, City Inspection, Predict AQI, Map View) with live sidebar filters.
- **Maps:** `india_aqi_map.html` — city-wise average AQI across all 26 cities.
- **Forecasts:** `monthly_forecast_top10.csv` — 60 rows of six-month-ahead AQI forecasts for ten cities.
- **Video Demo:** `https://youtu.be/y57es9DNrsM` — demonstration of the project in a 10 minute video.

Note: AI-assisted tools were used to support development. All analysis, code, and interpretations were reviewed and understood by the author.

8. REFERENCES

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